

REST API SPECIFICATION • v1.0

AsanShipping Developer API Documentation

Enterprise Integration Manual for Custom Storefronts, Mobile Apps, Inventory Ledgers, Order Ingestion, Courier Milestone Tracking & Webhook Subscriptions.

ENVIRONMENT BASE URLS

`https://merchant-api.asanshipping.com/api/v1`

PROTOCOL & AUTH

HTTPS / Bearer API Key

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1. System Overview & Architecture

The AsanShipping Developer API enables merchants and technology partners to programmatically access catalog, inventory ledger, canonical order placement, courier tracking, and real-time webhook subscriptions.

Enterprise Multi-Tenancy & Isolation: Every API key is strictly scoped to a single merchant organization. Cross-tenant access is prohibited at the database engine level.

Base URLs

Environment	Base URL	Purpose
Live Production	<code>https://merchant-api.asanshipping.com/api/v1</code>	Production store integrations & live order processing
Local / Sandbox	<code>http://localhost:5000/api/v1</code>	Local staging, CI/CD pipelines & test suites

Standard Request & Response Headers

Header Name	Type	Description
<code>Authorization</code>	string	Bearer authentication token: <code>Bearer as_live_...</code>
<code>X-API-Key</code>	string	Alternative header to pass secret or storefront API keys
<code>Idempotency-Key</code>	string (UUIDv4)	Guarantees atomic, non-duplicate execution on mutating POST/PATCH endpoints
<code>X-Request-Id</code>	string	Unique telemetry identifier returned in every response header for tracing
<code>X-Cache-Lookup</code>	string	Set to <code>IDEMPOTENT_HIT</code> when returning a cached 24h replay response

Authentication & Permission Scopes

API keys are created in the Merchant Portal under **Settings → Developer & API**.

Scope Key	Description
<code>products:read</code>	View catalog items, prices, variants, and image URLs.
<code>products:write</code>	Create/edit products, manage SKUs, and upload WebP images to Backblaze B2.
<code>inventory:read</code>	Query real-time stock balances across warehouses.
<code>inventory:write</code>	Submit authoritative adjustments (restocks/damages) and transfers.
<code>orders:read</code>	Query canonical orders, customer contacts, and consignments.
<code>orders:write</code>	Ingest orders with automated phone normalization and destination checks.
<code>orders:cancel</code>	Cancel unbooked orders with automatic inventory restocking.
<code>tracking:read</code>	Public and authenticated parcel milestone tracking timeline.

Scope Key	Description
webhooks:manage	Register, inspect, and test webhook subscriptions.

2. Reliability, Idempotency & Errors

Idempotency Protocol (24–Hour Window)

To eliminate duplicate charges and repeated order creation during network dropouts, AsanShipping implements an enterprise Idempotency Protocol on mutating endpoints (`POST /orders` , `POST /inventory/adjustments` , `POST /products`).

How to use: Send a unique UUIDv4 in the `Idempotency-Key` header. If a network timeout occurs and your client resends the identical request within 24 hours, the server returns the cached response with the `X-Cache-Lookup: IDEMPOTENT_HIT` header without executing duplicate side-effects.

Rate Limits

Environment Tier	Rate Limit	Burst Allowance	429 Penalty Window
Live Production (<code>as_live...</code>)	120 requests / min	30 requests	Check <code>Retry-After</code> header
Sandbox Testing (<code>as_test...</code>)	60 requests / min	15 requests	Check <code>Retry-After</code> header

Unified Error Protocol

All client and server errors adhere to the standard structured error schema:

```
{
  "success": false,
  "error": {
    "code": "INSUFFICIENT_STOCK",
    "message": "Requested quantity (5) exceeds available stock (2) for SKU 'TS-BLK-M'",
    "details": { "sku": "TS-BLK-M", "requested": 5, "available": 2 },
    "timestamp": "2026-09-26T14:00:00.000Z"
  },
  "meta": {
    "requestId": "req_01J8EF89X...",
    "documentation": "https://asanshipping.com/developers#rate-limits"
  }
}
```

Error Code Directory

Status	Error Code	Meaning & Resolution
401	<code>UNAUTHORIZED</code>	Missing or invalid API key in Authorization header.
403	<code>FORBIDDEN_INSUFFICIENT_SCOPE</code>	API key lacks the required permission scope.
400	<code>INVALID_IDEMPOTENCY_KEY</code>	Malformed or empty Idempotency-Key header on mutating route.
409	<code>IDEMPOTENCY_PAYLOAD_MISMATCH</code>	Reused Idempotency-Key with different request payload.
429	<code>RATE_LIMIT_EXCEEDED</code>	Exceeded allowed requests per minute.
422	<code>INSUFFICIENT_STOCK</code>	Requested quantity exceeds available ledger balance.

Status	Error Code	Meaning & Resolution
400	ORDER_ALREADY_DISPATCHED	Cannot cancel or modify an order already booked with courier.

3. Store Profile & Catalog Products

GET /store

Scope: Valid API Key

Retrieve merchant profile, operational currency, timezone, and warehouse fulfillments.

```
// Response 200 OK
{
  "success": true,
  "data": {
    "id": "66ce3698ff8e3e449a622b10",
    "name": "Luxe Couture Official",
    "currency": "PKR",
    "timezone": "Asia/Karachi",
    "warehouses": [
      { "id": "wh-khi-01", "name": "Karachi Central Hub", "city": "Karachi", "isDefault": true }
    ],
    "supportedPaymentMethods": ["COD", "Prepaid"],
    "supportedCouriers": ["TCS", "Trax", "PostEx", "Leopards", "M&P"]
  }
}
```

GET /products

Scope: products:read

Query paginated list of catalog products. Supports query parameters: `page`, `limit`, `search`, `sku`, `status`, `stockStatus`.

POST /products

Scope: products:write [Idempotent]

Create a catalog product with variant pricing, initial inventory allocation, and dimensional weight.

```
// Request Body JSON
{
  "sku": "HD-EMR-XL",
  "title": "Emerald Heavyweight Hoodie",
  "price": 4999,
  "compareAtPrice": 5999,
  "weightGrams": 650,
  "inventory": { "warehouseId": "wh-khi-01", "quantity": 40 }
}
```

POST /products/:id/images

Scope: products:write

Upload a product image (JPG/PNG/WebP, up to 10MB). Automatically compressed and stored on Backblaze B2.

4. Authoritative Inventory & Canonical Orders

GET `/inventory`

Scope: `inventory:read`

Retrieve authoritative on-hand, reserved, and available stock quantities per warehouse node.

POST `/inventory/adjustments`

Scope: `inventory:write` [Idempotent]

Submit atomic delta adjustments (+50 restock, -2 damage) with audit trail.

```
{
  "sku": "TS-BLK-M",
  "warehouseId": "wh-khi-01",
  "quantityDelta": 25,
  "reason": "RESTOCK",
  "referenceId": "PO-2026-992"
}
```

POST `/orders`

Scope: `orders:write` [Idempotent]

Submit a canonical customer order. Automatically validates destination city & subarea against AsanShipping directory, normalizes mobile numbers to E.164 (+92), reserves inventory, and logs an idempotent transaction.

```
// Request Body JSON
{
  "orderNumber": "WEB-2026-9041",
  "customer": { "name": "Bilal Ahmed", "phone": "03001234567", "email": "bilal@example.com" },
  "shippingAddress": { "address1": "House 42, Street 5, DHA Phase 6", "city": "Lahore", "subarea": "D
HA Phase 6" },
  "items": [{ "sku": "TS-BLK-M", "quantity": 2, "unitPrice": 2499 }],
  "paymentMethod": "COD",
  "totalAmount": 5248
}
```

POST `/orders/:id/cancel`

Scope: `orders:cancel`

Cancel an unbooked order and automatically restock reserved items back to warehouse ledger.

5. Delivery Locations & Address Reader Tool

GET **/locations/cities**

Scope: locations:read / Storefront

Access all available Pakistani delivery cities, provinces, logistics tiers, and sub-area coverage zones. Ideal for checkout city dropdowns and destination validation.

```
// Query Parameters: ?province=Punjab&search=Lahore&includeAreas=true
{
  "success": true,
  "data": [
    {
      "name": "Lahore",
      "province": "Punjab",
      "tier": "Tier 1",
      "areasCount": 42,
      "areas": ["Johar Town", "Gulberg", "DHA Phase 5", "Model Town", "Bahria Town"]
    },
    {
      "name": "Karachi",
      "province": "Sindh",
      "tier": "Tier 1",
      "areasCount": 56,
      "areas": ["Clifton", "DHA Phase 6", "Gulshan-e-Iqbal", "North Nazimabad"]
    }
  ]
}
```

POST /tools/parse-address

Scope: tools:execute / orders:write

Intelligently parse unstructured customer text into normalized Pakistani phone numbers (+923...), standardized delivery cities, sub-areas, and clean street addresses with confidence scoring. Supports single strings and high-speed bulk arrays (up to 100 items per request).

```
// Request Body JSON (Single or Bulk)
{
  "addresses": [
    "Ali Khan, 0300-1234567, Flat 4B, Johar Town, Lhr",
    "Fatima Tariq, +92 321 9876543, House 12, Gulshan-e-Iqbal, Karachi"
  ]
}

// Response JSON
{
  "success": true,
  "data": {
    "results": [
      {
        "originalText": "Ali Khan, 0300-1234567, Flat 4B, Johar Town, Lhr",
        "detectedPhone": "03001234567",
        "phoneNormalized": "+923001234567",
        "detectedCity": "Lahore",
        "detectedProvince": "Punjab",
        "detectedSubArea": "Johar Town",
        "cleanedStreetAddress": "Ali Khan, Flat 4B",
        "confidence": { "city": 98, "subArea": 100, "phone": 100, "overall": 99 }
      }
    ],
    "totalProcessed": 1
  }
}
```

6. Public Tracking & Webhooks

GET `/tracking/:trackingNumber`

Scope: tracking:read / Public

Public lookup endpoint returning the complete chronological journey milestones and COD balance.

```
{
  "success": true,
  "data": {
    "trackingNumber": "ASN-PK-8921849",
    "courier": "Trax Logistics",
    "status": "OUT_FOR_DELIVERY",
    "originCity": "Karachi",
    "destinationCity": "Lahore",
    "codAmount": 5248,
    "milestones": [
      { "status": "BOOKED", "location": "Karachi Hub", "timestamp": "2026-09-25T14:20:00Z" },
      { "status": "IN_TRANSIT", "location": "Lahore Sort Facility", "timestamp": "2026-09-26T06:45:00Z" },
      { "status": "OUT_FOR_DELIVERY", "location": "DHA Lahore Branch", "timestamp": "2026-09-26T10:15:00Z" }
    ]
  }
}
```

Webhooks & HMAC-SHA256 Signature Verification

Every webhook payload sent to your server contains the header `X-Asan-Signature: sha256={timestamp}.{hmacHex}`.

Node.js Verification Implementation

```
import crypto from 'crypto';

export function verifyAsanWebhook(rawBodyBuffer, signatureHeader, webhookSecret) {
  const parts = signatureHeader.split('.');
  const timestamp = parts[0].replace('sha256=', '');
  const signature = parts[1];

  // Prevent replay attacks (reject payloads older than 5 minutes)
  if (parseInt(timestamp) < Math.floor(Date.now() / 1000) - 300) {
    throw new Error('Webhook timestamp expired');
  }

  const expected = crypto
    .createHmac('sha256', webhookSecret)
    .update(`${timestamp}.${rawBodyBuffer.toString('utf8')}`)
    .digest('hex');

  return crypto.timingSafeEqual(Buffer.from(signature, 'hex'), Buffer.from(expected, 'hex'));
}
```

Supported Webhook Topics

Topic	Description
<code>order.created</code>	Fired immediately after an order is accepted into the system.
<code>order.updated</code>	Fired when customer details, delivery address, or consignments are updated.
<code>order.cancelled</code>	Fired when an order is cancelled and items are returned to inventory.
<code>tracking.milestone_updated</code>	Fired when courier pushes a new tracking scan.
<code>inventory.low_stock</code>	Fired when stock drops below safety reorder threshold.